Chapter 11 • Loci and Construction

Solved Exercise 11.1

1(i). Construct AB=5 cm, BC=6 cm, AC=7 cm; verify concurrent perpendicular bisectors.

Draw BC=6 cm. With centre B draw radius 5 cm and with centre C radius 7 cm; their intersection is A. Join AB and AC.
Construct the perpendicular bisectors of AB, BC and CA. They meet at one point O, the circumcentre.
Check: OA=OB=OC; the triangle is acute, so O lies inside.

1(ii). Construct AB=7.1 cm, ∠B=135°, BC=6.5 cm; verify concurrent perpendicular bisectors.

Draw BC=6.5 cm. Construct a 135° ray at B and mark BA=7.1 cm. Join AC.
Construct all three side perpendicular bisectors; they meet at O.
Because ∠B is obtuse, the circumcentre lies outside the triangle, but concurrency still holds.

2(i). Construct LM=4.9 cm, ∠L=51°, ∠M=38°; verify concurrent medians.

Draw LM=4.9 cm; construct rays making 51° at L and 38° at M on the same side. Their intersection is N. The third angle is 91°.
Find the midpoint of each side using its perpendicular bisector. Join each midpoint to the opposite vertex.
The three medians meet at centroid G, which divides each median in ratio 2:1 from the vertex.
Medians are concurrent at centroid G.

2(ii). Construct MN=4.8 cm, ∠N=30°, LM=8.1 cm; verify concurrent medians.

Draw MN=4.8 cm and a 30° ray from N. Draw a circle centred at M with radius 8.1 cm; its intersection with the ray is L.
Join LM and LN, construct the three side midpoints and draw the medians.
Since LM>MN, the SSA data give one triangle on the chosen ray; the medians meet at G.

3(i). Construct AB=4.5 cm, ∠A=45°, AC=5.3 cm; verify concurrent angle bisectors.

Draw AB=4.5 cm, make a 45° ray at A and mark AC=5.3 cm. Join BC.
Bisect ∠A, ∠B and ∠C with compass arcs. The bisectors meet at I, the incentre.
A circle centred at I and perpendicular-distance from I to a side touches all three sides.
Angle bisectors concurrent at incentre I.

3(ii). Construct AB=6 cm, ∠A=150°, ∠B=60°; verify angle-bisector concurrency.

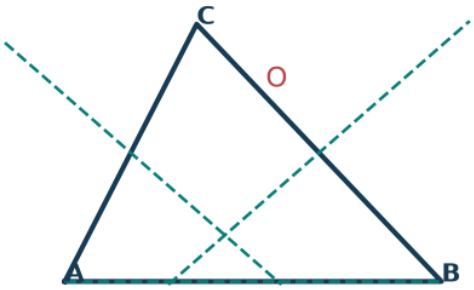
The two given angles already total 150°+60°=210°, exceeding the triangle angle sum of 180°.

4. For DE=4.8 cm, EF=4 cm and included ∠E=45°, draw the altitudes and locate the orthocentre.

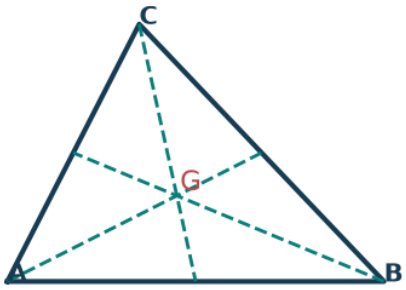
Draw DE=4.8 cm, construct a 45° ray at E and mark EF=4 cm. Join DF.
From F drop a perpendicular to DE; from D drop a perpendicular to EF; from E drop a perpendicular to DF. Their common intersection is H, the orthocentre.
Using E=(0,0), D=(4.8,0), F=(2.828,2.828), the altitudes intersect approximately at (2.828,1.972), inside this acute triangle.

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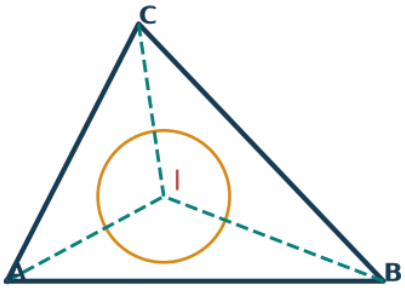
Perpendicular bisectors → circumcentre O



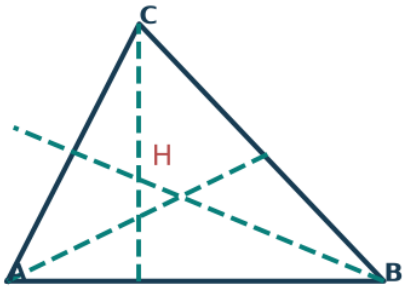
Medians → centroid G



Angle bisectors → incentre I



Altitudes → orthocentre H



Schematic construction guides — not to scale

5(i). Construct ΔBCD with BC=5 cm, ∠B=62°, CD=4.7 cm; investigate the ambiguous case.

Draw BC=5 cm and a 62° ray from B. A circle centred at C with radius 4.7 cm cuts that ray at two possible positions D_1 and D_2.

The perpendicular distance from C to the ray is $h = 5\sin 62° \approx 4.415$ cm. Since $h < 4.7 < 5$, two triangles exist.

5(ii). ΔKLM is printed with LM=6 cm, ∠M=42° and undefined LN=5 cm; use the intended SSA side LK=5 cm.

Draw LM=6 cm and a 42° ray from M. A circle centred at L with radius 5 cm cuts the ray at two possible positions K_1 and K_2.

Here $h = 6\sin 42° \approx 4.015$ cm, and $h < 5 < 6$, so two triangles exist under this intended reading.